

THE RECTUM (Surgery)

Anatomy

- ~15 cm; lower/middle/upper thirds. Blood: **superior, middle, inferior rectal vessels**. **Lymphatic drainage upward along superior rectal vessels** → **para-aortic nodes** (lower rectum → lateral pelvic walls).

Clinical features of rectal disease

- **Symptoms: fresh PR bleeding, altered bowel habit (loose stool), mucus discharge, tenesmus, prolapse, proctalgia.** (Inflammatory = young; tumours = middle-aged/elderly.)
- **Exam:** left lateral (Sims) position; inspect (fissure/fistula/prolapse); **DRE** (sphincter tone, tumours lower/middle third, check finger for mucus/pus/blood); proctoscopy; sigmoidoscopy.

Rectal prolapse

- **Mucosal prolapse:** mucosa + submucosa protrude 1–4 cm. Infants (undeveloped sacral curve), children (post-diarrhoea, CF, Hirschsprung), adults (with **3rd-degree haemorrhoids = mucohaemorrhoidal**). Tx: digital repositioning (infants), phenol injection/banding, excision/Delorme's.
- **Full-thickness (procidentia):** all layers, **>4 cm (often 10–15cm)**, patulous gaping sphincter. **Women 6x > men; 50% faecal incontinence.** Complications: ulceration, bleeding, incarceration/strangulation.
- **Treatment (surgery): abdominal rectopexy** (lower recurrence 10%; laparoscopic anterior mesh rectopexy favoured) vs **perineal** (frail/elderly): **Delorme's** (short segment, 30% recurrence), **Altemeier's** (incarcerated/strangulated; poor bowel control).

Proctitis

- Ulcerative proctocolitis, Crohn's, radiation, infections (**C. difficile, bacillary/amoebic dysentery, TB, gonococcal, LGV**), bilharziasis, AIDS.

Rectal polyps

- **Remove all for histology + exclude carcinoma.** Described by appearance (pedunculated/sessile/flat) + histology (tubular/villous/tubulovillous).
- Types: hyperplastic, **tubular/villous/serrated adenomas**, FAP, inflammatory pseudopolyps, juvenile.
- **Polyps >1 cm = 10% malignancy. Villous = more malignant change than tubular. EMR <2cm, ESD >2cm.** All adenomas removed; colonoscopy for synchronous polyps.

Carcinoma (colorectal / rectal)

- **CRC = 2nd commonest malignancy** (2nd in women, 3rd in men); **4th cancer death** (after lung, gastric, liver). **Rectum = most frequent site (~1/3).** Most >70 yrs.
- **Pathogenesis: adenoma-carcinoma sequence.** 80% sporadic. IBD ↑risk; family history (2+ relatives = 2–3x, ~20% cases); **HNPCC/Lynch (3%)**, FAP, Gardner. Genes: **Wnt/APC (commonest)**, TP53, KRAS, PI3K, PTEN.
- **Clinical:** presentation >55 yrs. Early — **PR bleeding, tenesmus (spurious diarrhoea, bloody slime), early morning bloody diarrhoea.** Late — colicky pain (obstruction), invasion (prostate/bladder/sacral plexus), weight loss (metastatic).
- **Exam:** DRE if within 7–8 cm of anal verge; vaginal exam (posterior wall).

- **Diagnosis: colonoscopy + biopsy**; CT TAP (staging); **MRI rectum (local — mesorectum, fascia)**; **CEA** (baseline/follow-up); PET (equivocal mets).
- **Spread: local = circumferential** (most important for management); lymphatic upward (superior rectal → para-aortic; **lateral pelvic in low rectal ~20%**); venous → **liver 34%**, **lungs 22%**, **adrenals 11%**; transperitoneal.

Staging

- **Dukes: A** (rectal wall, >90% 5-yr), **B** (extrarectal, no nodes, 70%), **C** (regional nodes — C1 pararectal, C2 to vessel origin, 40%), **D** (widespread mets — not original Dukes).
- **TNM (international standard)**: T1 submucosa → T2 muscularis → T3 perirectal fat → T4 (a peritoneum, b adjacent organs); N1 (1–3 nodes), N2 (≥4); M1 (distant).
- **Grading**: well/moderate/poor differentiated; **signet ring = poor prognosis**; poor features = vascular/perineural invasion, infiltrating margin, tumour budding.

Treatment

- **MDT** assessment. **Radical excision of rectum + mesorectum + nodes (TME)**. Liver mets don't preclude cure (>1/3 long-term survival).
- Operations: **anterior resection, abdominoperineal excision (APR), Hartmann's, transanal TME, local excision**; endoluminal stenting/palliative colostomy/pelvic exenteration/liver resection.

High-yield one-liners

- **Rectal symptoms: bleeding, tenesmus, mucus, altered habit; DRE essential.**
- **Full-thickness prolapse = women 6×, procidentia; abdominal rectopexy (low recurrence) vs perineal (frail).**
- **Polyps >1cm = 10% malignant; villous worse; remove all + colonoscopy.**
- **CRC = adenoma-carcinoma sequence (APC); rectum commonest site; circumferential local spread dictates management.**
- **Dukes A-D / TNM; liver mets commonest; MRI for rectal local staging, CEA follow-up.**
- **TME + anterior resection / APR; signet ring = poor prognosis.**

THE RECTUM (Surgery) — Revision Table

Bold = high-yield. Dr Dara Mohammed.

CLINICAL & PROLAPSE

Topic	Key points
Symptoms	Fresh PR bleeding, altered habit, mucus, tenesmus, prolapse, proctalgia ; DRE essential
Mucosal prolapse	1–4cm; adults with 3rd-degree haemorrhoids (mucohaemorrhoidal) ; phenol/banding
Full-thickness (proctidentia)	All layers, >4cm; women 6x; 50% incontinence; abdominal rectopexy (recurrence 10%) vs perineal (Delorme's/Altemeier's, frail)
Proctitis	UC, Crohn's, radiation, C. diff , amoebic, TB , LGV , bilharzia

POLYPS & CARCINOMA

Topic	Key points
Polyps	>1cm = 10% malignancy; villous > tubular ; EMR <2cm, ESD >2cm; remove all + colonoscopy
CRC overview	2nd commonest cancer; rectum most frequent site (1/3) ; adenoma-carcinoma sequence (APC, TP53, KRAS)
Risk	IBD, family history (HNPCC/Lynch 3%, FAP)
Clinical	Bleeding, tenesmus (spurious diarrhoea, bloody slime) ; late = pain/weight loss
Diagnosis	Colonoscopy + biopsy; MRI rectum (local); CEA; CT TAP staging
Spread	Circumferential local (dictates management) ; lymphatic upward; venous → liver 34%, lung 22%

STAGING & TREATMENT

Topic	Key points
Dukes	A (wall, >90%), B (extrarectal, 70%), C (nodes, 40%), D (mets)
TNM	T1 submucosa → T4 organs; N1 (1–3), N2 (≥4 nodes); M1 distant
Grading	Well/mod/poor; signet ring = poor ; vascular/perineural invasion
Treatment	MDT; TME (mesorectal excision) + radical resection ; anterior resection/APR/Hartmann's; liver mets resectable